Docker allows you to define a *restart policy* that says how it should behave when the container terminates.[21] By setting this to on-failure:

```
deploy:
  restart_policy:
    condition: on-failure
```

Swarm will now automatically restart our Rails app were it to crash. Here's a good article with more details.[22]

Multi-stage Builds

Large Docker images are slower to push and pull. As you become more experienced with Docker, you'll want to find ways to make your images. Since version 17.05, Docker has had a feature called multi-stage builds[23]—this lets you use multiple FROM statements in a single Dockerfile. Each new FROM is considered a new stage, and starts as a fresh new image. However, the COPY instruction has been enhanced to let you copy files from *earlier stages*.

The most obvious use case is where you need a lot of development tools that produce a final artifact. Think of a static site generator like Jekyll[24] or Middleman[25]—you need various tools to develop and generate the site, but once the static files are generated, they're the only thing needed to run the site. Multi-stage builds let you create an initial stage that generates the site, and a separate, final stage that copies those files into a clean web server image. The same goes for compiled languages like Go where, typically, the only thing you need to include in your final image is the compiled binary.

In the case of our Rails app, a quick win could be copying the precompiled assets into a final image, avoiding the need for all the JavaScript dependencies. There are other ways to save space if you think creatively and see what other people are doing.

Docker Stats

Often, especially in production, it's useful to have a quick way to find out metrics about the resources being used. The Docker docs[26] provide some useful information on various metrics you can check out.

21. https://docs.docker.com/config/containers/start-containers-automatically/#use-a-restart-policy
22. https://blog.codeship.com/ensuring-containers-are-always-running-with-dockers-restart-policy/
23. https://docs.docker.com/develop/develop-images/multistage-build/
24. https://jekyllrb.com
25. https://middlemanapp.com
26. https://docs.docker.com/config/containers/runmetrics/